

Multiple Choice Quiz

Understanding DNS Architecture and Mechanics

Total Questions: 10

Time Allowed: 20 minutes

Instructions: Choose the **best** answer for each question. Each question carries equal marks.

Question 1

What is the fundamental purpose of the Domain Name System (DNS)?

- A. To assign permanent IP addresses to all internet-connected devices
 - B. To translate human-readable domain names into machine-routable IP addresses
 - C. To encrypt all internet traffic between clients and servers
 - D. To store website content and deliver it to users
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Question 2

How is the DNS namespace organized, and what are its character constraints?

- A. It uses a flat structure where every name is stored at the root level
 - B. Labels can be up to 255 characters long with no limit on the full name
 - C. It is organized as a linear list with case-sensitive labels
 - D. It is organized as a rooted tree where labels are limited to 63 characters and full names to 255 characters
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Question 3

What is the architectural distinction between a Domain and a Zone?

- A. A Domain is a logical subtree in the naming hierarchy, while a Zone is the unit of administrative delegation and responsibility
 - B. A Zone is a logical subtree, while a Domain is the unit of administrative control
 - C. There is no difference — both terms refer to the same concept
 - D. A Domain is controlled by ICANN, while a Zone is controlled by individual users
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Question 4

How do Primary and Secondary servers interact to ensure data availability?

- A. The Secondary server is the only server that can accept updates from clients
 - B. Both servers maintain completely independent databases with no synchronization
 - C. The Primary server holds the master copy, and the Secondary server obtains updates through zone transfers
 - D. The Secondary server sends updates to the Primary server using multicast
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Question 5

Explain the “Principle of Locality” as it relates to DNS scalability.

- A. Every DNS query must be sent to the root servers to ensure global consistency
 - B. Most DNS queries are answered from local caches at ISPs or resolvers, reducing global traffic
 - C. DNS performance improves when all queries are sent to a single central server
 - D. Locality has no impact on DNS scalability
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Question 6

What are the primary differences between Recursive and Iterative name resolution?

- A. Recursive resolution places a heavy burden on the client, while Iterative resolution is handled entirely by the resolver
 - B. Both methods require the same number of network round trips
 - C. Iterative resolution is always faster than Recursive resolution
 - D. In Recursive resolution the resolver handles the full query chain (light client burden), while Iterative resolution requires the client to follow multiple referrals (heavy client burden)
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Question 7

How does the Start of Authority (SOA) record function within a DNS zone?

- A. It defines the administrative rules and validity parameters for how zone data should be handled and refreshed
 - B. It stores the IP addresses of all hosts within the zone
 - C. It is used only for email routing within the domain
 - D. It encrypts all DNS traffic leaving the zone
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Question 8

What was the original 1980s constraint of DNS that necessitated modern updates?

- A. DNS was originally designed with strong encryption and authentication mechanisms
 - B. The protocol only supported IPv6 addresses from the beginning
 - C. DNS was built on small 512-byte UDP packets with no encryption or strong identity verification
 - D. The original DNS had built-in support for DNSSEC and DoH
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Question 9

Describe the specific roles of the Zone-Signing Key (ZSK) and Key-Signing Key (KSK) in DNSSEC.

- A. The KSK signs resource records while the ZSK signs the KSK
 - B. The ZSK signs the resource records within a zone, while the KSK signs the ZSK to create a chain of trust up to the root
 - C. Both keys are used interchangeably for signing all records
 - D. The ZSK and KSK are only used for encrypting DNS traffic, not for signing
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Question 10

What is the “Centralization Dilemma” regarding modern external resolvers?

- A. Using external resolvers makes the DNS system more decentralized
 - B. External resolvers are slower than local ISP resolvers in all cases
 - C. The dilemma has been completely solved by the introduction of DNSSEC
 - D. While external “Big Tech” resolvers offer speed through massive caching, they concentrate naming data and threaten the decentralized nature of DNS
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Answer Key

Question	Correct Answer	Explanation
1	B	DNS translates human-readable names (e.g., www.vu.nl) into machine-routable IP addresses (e.g., 130.37.20.20).
2	D	DNS uses a rooted tree structure. Labels are limited to 63 characters; full names

Question	Correct Answer	Explanation
3	A	are limited to 255 characters. A Domain is a logical subtree; a Zone is the administrative unit of delegation.
4	C	The Primary holds the master copy; the Secondary synchronizes via zone transfers .
5	B	The Principle of Locality means most queries are served from local caches, improving scalability.
6	D	Recursive = resolver does all the work (light client). Iterative = client follows referrals (heavy client).
7	A	The SOA record defines administrative rules and refresh/validity parameters for the zone.
8	C	Original DNS used small unencrypted UDP packets with weak security, leading to modern extensions (DNSSEC, DoT, DoH).
9	B	ZSK signs zone records; KSK signs the ZSK, creating a cryptographic chain of trust to the root.
10	D	Outsourcing to large external resolvers improves performance but centralizes DNS

Question	Correct Answer	Explanation
		data, threatening decentralization.

End of Quiz

Correct Answer Distribution: A = 2 | B = 3 | C = 2 | D = 3

This quiz tests conceptual understanding of DNS architecture based on the lecture materials.